

Automatic USB-Controlled Power Switch

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Almost all electronic gadgets now come with one or more USB ports. You can use the USB port as an automatic switch to

control devices powered by mains. Here we present a circuit that can control mains power of an LED TV and a DTH set-top box (STB).

Generally, mains power for TV and STB are derived from a common

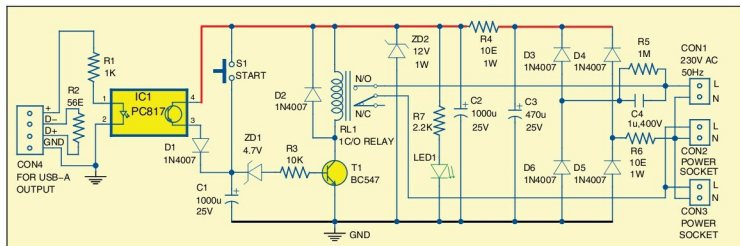


Fig. 1: Circuit diagram of the automatic USB-controlled power switch

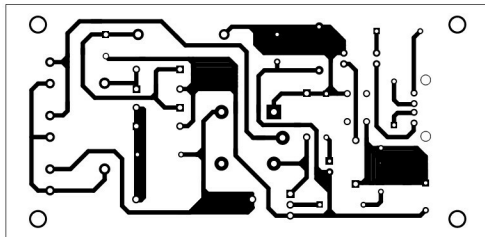


Fig. 2: Actual-size PCB of the automatic USB-controlled power switch

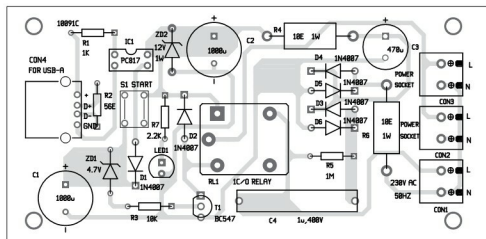


Fig. 3: Component layout of the PCB

switchboard. When you switch off the power of the TV using a remote control or sleep timer in the TV, the switched-mode power supply (SMPS) of the TV continues to draw a small amount of power from mains and the STB does not get switched off. This drawback can be overcome using this circuit. Besides, it protects the devices from harmful initial surges when power comes back after a power failure.

Circuit and working

Circuit diagram of the automatic USB-controlled power switch is shown in Fig. 1. It is built around optocoupler PC817 (IC1), transistor BC547 (T1), 12V 1C/O relay (RL1), six rectifier diodes 1N4007 (D1 through D6) and a few other components. USB-A extender cables are used to connect the USB ports of the TV and of the STB to the circuit.

230V AC mains power supply connector CON1 is connected between N/O contacts of relay RL1 and neutral line N. When switch S1 is pressed, it allows positive voltage to